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Japanese (Professional working, JLPT N1 121/180) · English (Professional, TOEFL 97/120) · Chinese (Native)

🎓 Education

The University of Tokyo 東京大学

Oct. 2025 – Sep. 2027 (Expected)

M.S. Candidate, Information & Communication Engineering

Tokyo, Japan

- Suzumura Laboratory; pursuing research on **Vision-Language-Action (VLA)** models and human-robot collaboration benchmarks for scientific laboratory environments.

Shanghai University 上海大学

Sep. 2021 – Jul. 2025

B.E., Electronic Information Engineering

Shanghai, China

🔧 Projects

KotobaLab

Apr. 2026 – Present

Personal Project: SwiftUI, GRDB 7, SQLite, SwiftData, Swift Testing, iOS 18+

Tokyo, Japan

Local-first Japanese dictionary app built with SwiftUI, MVVM, and a Clean Architecture-inspired layered design.

Currently in local development.

- Built a local-first dictionary app with a **293K-entry, 52 MB SQLite** database generated by a Python pipeline, bundled as an app resource and queried through GRDB DatabaseQueue.
- Optimized prefix search with `PRAGMA case_sensitive_like=ON`, improving benchmarked queries from ~16 ms table scans to **~0.03 ms** multi-index lookups.
- Designed a four-layer architecture (AppDependencies → Scene → Store → View) following **MVVM** and **Clean Architecture** principles; applied the **Repository pattern** with protocol-based abstractions and dependency injection to decouple Domain logic from UI and storage frameworks.
- Combined GRDB-managed SQLite for read-heavy reference data with SwiftData for user state; wrote 13 unit tests with Swift Testing covering domain use cases and repository behavior.
- Roadmap: SQLite FTS5 full-text search, actor-isolated repositories under Swift Concurrency, TestFlight beta.
- Github: [📁 shiinayane/KotobaLab](https://github.com/shiinayane/KotobaLab).

LabMate

Dec. 2025 – Present

Suzumura Lab: Python, Isaac Sim, VLA Models

Tokyo, Japan

Simulation-first benchmark for human-robot collaboration in scientific laboratory environments.

- Extending the **LabUtopia** simulator into a benchmark for natural-language human-robot collaboration in scientific laboratories, focusing on clarification, safety-aware refusal, grounding, and structured action logging.
- Designing evaluation framework to compare planner architectures (**LLM**, **VLA**, and hybrid systems) on multi-step laboratory protocols; sim-first phase planned for 6+ months before hardware integration.
- Coordinating with 3 procured robotic platforms for future hardware integration; prioritizing rigorous evaluation methodology over demonstration-driven research.

👤 Experience

GlucoPI

Feb. 2025 – Jun. 2025

B.E. Graduation Project, Advisor: Qi Zhang, PhD, Professor

Shanghai, China

WeChat Mini Program for diabetes management, doctor-patient communication, and glucose prediction.

- Built an end-to-end diabetes self-tracking app with WeChat Mini Program frontend, FastAPI backend, MySQL/MongoDB storage, and WebSocket doctor-patient chat.
- Implemented login and permissions, glucose / diet / insulin records, doctor-patient pairing, real-time messaging, and trend charts.
- Added short-horizon glucose prediction in PyTorch by reproducing the GluPred pipeline and running experiments on **OhioT1DM**.

- Integrated a lightweight LLM Q&A feature for non-diagnostic health guidance.
- Github: [shiinayane/glucopi](https://github.com/shiinayane/glucopi).

Arrived or Not

Dec. 2023 – Dec. 2024

Role: Co-Leader, **Advisor:** Qi Zhang, PhD, Professor

Shanghai, China

Machine-vision teaching platform for classroom attendance and course management.

- Implemented real-time face-recognition attendance with RetinaFace detection and **ArcFace** embedding-based identity matching.
- Built teacher / student mobile workflows for class setup, attendance records, and basic engagement statistics.
- Developed a Flutter client and FastAPI service with a clear frontend-backend split.
- Github: [shiinayane/Arrived-or-Not-Frontend](https://github.com/shiinayane/Arrived-or-Not-Frontend) and [shiinayane/Arrived-or-Not-Backend](https://github.com/shiinayane/Arrived-or-Not-Backend).

💡 Interests

Product-First Engineering: Building products that solve real problems; treating iOS, backend, and full-stack as means rather than ends.

iOS Native Architecture: Local-first design with clean Domain / UI separation; currently focused on shipping native iOS apps as the most direct path to product impact.

Research-to-Product: Bridging VLA / LLM research from graduate work to consumer mobile experiences.

⚙️ Skills

iOS / Apple Platforms: Swift, SwiftUI, GRDB, SQLite, SwiftData, Observation, Swift Concurrency, Swift Testing

Architecture & Design: MVVM, Clean Architecture, Repository Pattern, Dependency Injection, Protocol-Oriented Programming

Backend / Web: Python (FastAPI), JavaScript

ML / Research: PyTorch, VLA Models, Isaac Sim

Tools / Platforms: Git, Docker, Xcode, VS Code, macOS, Linux